Teacher answer key: Primary 6, Semester 2 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 6 Standard content in the Mathematics Syllabus, Primary (2021 edition, pages 43 and 44). The outcome wording in this booklet is a summary in our own words, not the syllabus text; where the summary gave no outcome number the key gives none. The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 3 and 4)

Strand	Pages
Measurement and Geometry - Area and Volume	S2-01, S2-02, S2-03, S2-04, S2-05, S2-06
Measurement and Geometry - Geometry	S2-07, S2-08, S2-09
Statistics	S2-10, S2-11
Measurement and Geometry - Area and Volume; Statistics	S2-12
Revision of earlier years	S2-13, S2-14, S2-15, S2-16, S2-17, S2-18, S2-19
Number and Algebra - mixed Primary 6	S2-20

The page code in the footer, for example S2-07 (P6-27), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S2-01 (P6-21) Shaded Shapes 1 star

Topic: Area of composite figures

Syllabus outcomes: Area and Volume 1.3. Notes on the tag: 1.3 (composite figures made of squares, rectangles, triangles, semicircles and quarter circles)
Puzzle type: Code-breaker (composite areas). How the pupil checks: Code word (6 letters).
Answers: 1) 1036 cm²; 2) 273 cm²; 3) 42 cm²; 4) 476 cm²; 5) 64 cm²; 6) 56 cm²
Check result: Code word: SUNDAY
Teacher note: Six figures: a rectangle with a semicircle on one end, a square with a semicircle on one side, a square with a quarter circle cut away (hatched), a square with a semicircle cut out (hatched), and two rectangles with a triangle on top. Sizes are chosen so that every answer is a whole number with π as 22/7. The decoy numbers are what a pupil gets by using a full circle instead of a half, by forgetting the half in the triangle, or by adding instead of subtracting. A box left blank gives no letter and the check cannot see it.

S2-02 (P6-22) Perimeter Pick 2 stars

Topic: Perimeter of figures with curved edges

Syllabus outcomes: Area and Volume 1.2, Area and Volume 1.3. Notes on the tag: 1.2, 1.3 (perimeter of a semicircle and a quarter circle; composite figures)
Puzzle type: Choice table (perimeters). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 100 cm; 2) 76 cm; 3) 54 cm; 4) 64 cm; 5) 142 cm; 6) 25 cm
Check result: Word: WALNUT
Teacher note: Six figures: a semicircle, a quarter circle, two rectangles with a semicircle on one end, a square with a semicircle on one side and a square with a quarter circle cut away (the hatched part). Each row has three answers. The wrong answers are what a pupil gets by leaving out the straight edges, by counting a dashed line that is inside the figure, or by using the whole circle. The perimeter of a semicircle includes its diameter. The letters of the right answers, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-03 (P6-23) Cuboid Corners 1 star

Topic: Volume: finding a missing length of a cuboid

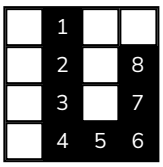
Syllabus outcomes: Area and Volume 2.1. Notes on the tag: 2.1 (finding a missing dimension of a cuboid given its volume and the other dimensions)
Puzzle type: Code-breaker (missing dimension). How the pupil checks: Code word (6 letters).
Answers: 1) h = 8 cm; 2) l = 14 cm; 3) b = 5 cm; 4) h = 11 cm; 5) b = 12 cm; 6) l = 15 cm
Check result: Code word: COUGAR
Teacher note: Six cuboids: three drawn with one edge marked by a question mark and three in words. The volume is given and two of the three edges; the pupil divides the volume by the product of the two edges. The decoy numbers are what a pupil gets by dividing the volume by only one edge, by multiplying the two edges, or by taking the volume less the product of the edges. A box left blank gives no letter and the check cannot see it.

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S2-04 (P6-24) Roots and Edges 1 star

Topic: Square roots, cube roots and the edge of a cube

Syllabus outcomes: Area and Volume 2.5, Area and Volume 2.2. Notes on the tag: 2.5 (square-root and cube-root signs); 2.2 (the edge of a cube from its volume)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box equal to 6.
Answers: Route (row, column from top-left): (1,2) $3 \times \sqrt[3]{8}$ -> (2,2) $\sqrt[3]{27} + 3$ -> (3,2) $\sqrt[3]{125} + 1$ -> (4,2) $2 \times \sqrt[3]{27}$ -> (4,3) $\sqrt{25} + 1$ -> (4,4) $\sqrt{49} - 1$ -> (3,4) $\sqrt[3]{8} + 4$ -> (2,4) $\sqrt[3]{343} - 1$
Check result: Route reaches FINISH through 8 boxes; every box on it equals 6.



Teacher note: A 4 by 4 grid of sixteen boxes: square roots, cube roots, a root with a number added, taken away or multiplied, the edge of a cube from its volume and the side of a square from its area. The route follows the boxes that equal 6. The wrong boxes include answers close to 6. Two extra boxes that equal 6 are dead ends. Counting the boxes shows a line that takes a shortcut.

S2-05 (P6-25) Base and Height 2 stars

Topic: Volume: base area, height and faces

Syllabus outcomes: Area and Volume 2.3, Area and Volume 2.4, Area and Volume 2.2. Notes on the tag: 2.3, 2.4, 2.2 (height of a cuboid from volume and base area; area of a face from volume and one dimension; edge of a cube)
Puzzle type: Matching (base area, height, face). How the pupil checks: Letters of the matching cards, read in the order of the items, spell a word (6 letters).
Answers: 1) card C; 2) card A; 3) card N; 4) card O; 5) card P; 6) card Y
Check result: Word: CANOPY
Teacher note: Six questions: the height from the volume and the base area, the base area from the volume and the height, the area of a face from the volume and one edge, the depth of water in a tank, the face of a cube from its volume, and the height from the volume and a base given by its two edges. Two cards are decoys. A card carries a number and a unit: lengths are in cm and areas in cm^2. The letters of the matching cards, read in the order of the items, spell the word. Every way of giving each item a different card, and every single changed letter, was tested against a very large word list: none spells another word. A box left blank gives no letter and the check cannot see it.

S2-06 (P6-26) Tank Stories 2 stars

Topic: Volume of water in a tank: litres and cm^3

Syllabus outcomes: Area and Volume 2.3, Revisits Primary 5 Volume. Notes on the tag: 2.3 (height from volume and base area); revisits Primary 5 Volume (litres and cm^3)
Puzzle type: Code-breaker (tanks). How the pupil checks: Code word (6 letters).
Answers: 1) 64; 2) 21; 3) 8; 4) 30; 5) 40; 6) 18
Check result: Code word: MAGNET
Teacher note: Six tank stories that need litres changed to cm^3 or back: the depth of water, the litres in a tank, half a tank, the height reached by water poured in, the height of a tank that is part full, and a full cube-shaped tank. The decoy numbers are what a pupil gets by forgetting to change litres to cm^3 (an answer ten times or a thousand times too big or small), by forgetting the fraction, or by using one edge instead of the base area. The relationship 1 litre = 1000 cm^3 is printed on the page. A box left blank gives no letter and the check cannot see it.

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S2-07 (P6-27) Angles in Figures 1 star

Topic: Unknown angles in composite figures

Syllabus outcomes: Geometry 1.1. Notes on the tag: 1.1 (finding unknown angles in composite figures involving squares, rectangles, triangles, parallelograms, rhombuses and trapeziums; no additional construction)
Puzzle type: Code-breaker (unknown angles). How the pupil checks: Code word (6 letters).
Answers: 1) 120°; 2) 85°; 3) 136°; 4) 22°; 5) 122°; 6) 80°
Check result: Code word: PEANUT
Teacher note: Six figures in one or two steps: a trapezium (the angles between the parallel sides add to 180 degrees), a triangle with a straight line (the angle sum of a triangle and angles on a straight line), an isosceles triangle with a straight line, and a rhombus with a diagonal (all four sides are equal, so a triangle is isosceles). The decoy numbers are what a pupil gets by using 180 minus the answer, by using the given angle, or by halving or doubling. The figures are drawn accurately so the pupil could measure, but the aim is to work the angle out. A box left blank gives no letter and the check cannot see it.

S2-08 (P6-28) More Angles 2 stars

Topic: Unknown angles in composite figures

Syllabus outcomes: Geometry 1.1. Notes on the tag: 1.1 (finding unknown angles in composite figures involving squares, rectangles, triangles, parallelograms, rhombuses and trapeziums; no additional construction)
Puzzle type: Code-breaker (unknown angles). How the pupil checks: Code word (6 letters).
Answers: 1) 36°; 2) 54°; 3) 64°; 4) 15°; 5) 75°; 6) 24°
Check result: Code word: SPONGE
Teacher note: Six figures that need two or three steps: a parallelogram with an isosceles triangle (opposite angles of a parallelogram are equal), an isosceles triangle where the exterior angle is given, a rhombus where the opposite angle is given, and a square with an equilateral triangle on one side, outside and inside. In the square figures the pupil finds an angle of a triangle whose two equal sides are the side of the square and the side of the triangle. The decoy numbers are what a pupil gets by using 180 minus the answer, by using the given angle, or by halving or doubling. The figures are drawn accurately, but the aim is to work the angle out. A box left blank gives no letter and the check cannot see it.

S2-09 (P6-29) Angle Match 1 star

Topic: Unknown angles in composite figures

Syllabus outcomes: Geometry 1.1. Notes on the tag: 1.1 (finding unknown angles in composite figures; no additional construction)
Puzzle type: Matching (angles). How the pupil checks: Letters of the matching cards, read in the order of the items, spell a word (6 letters).
Answers: 1) card S; 2) card T; 3) card R; 4) card O; 5) card N; 6) card G
Check result: Word: STRONG
Teacher note: Six figures (an isosceles triangle with a straight line, a rhombus with a diagonal, a trapezium, a triangle with a straight line, an isosceles triangle with the exterior angle given, and a parallelogram with an isosceles triangle) matched to eight cards of degrees. Two cards are decoys: 180 minus the answer to one figure. The letters of the matching cards, read in the order of the items, spell the word. Every way of giving each item a different card, and every single changed letter, was tested against a very large word list: none spells another word. The figures are drawn accurately but the aim is to work the angle out. A box left blank gives no letter and the check cannot see it.

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S2-10 (P6-30) Average 1 star

Topic: Average: total divided by the number of data

Syllabus outcomes: Statistics 1.1. Notes on the tag: 1.1 (average as total value ÷ number of data)
Puzzle type: Code-breaker (average). How the pupil checks: Code word (6 letters).
Answers: 1) 20; 2) 13; 3) 29; 4) 31; 5) 77; 6) 30
Check result: Code word: CURFEW
Teacher note: Six items: the average of four numbers, of three test marks, of five daily counts and of five box counts, and two items where the total and the number are given. In each list the total divides exactly by the number of items. The decoy numbers are what a pupil gets by dividing by the wrong number of items, by adding one too many or too few, or by taking the total. A box left blank gives no letter and the check cannot see it.

S2-11 (P6-31) Missing Value 2 stars

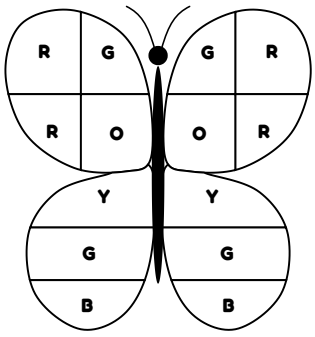
Topic: Average: total, number and average

Syllabus outcomes: Statistics 1.2. Notes on the tag: 1.2 (relationship between average, total value and number of data)
Puzzle type: Code-breaker (average, total and number). How the pupil checks: Code word (6 letters).
Answers: 1) 78; 2) 21; 3) 35; 4) 15; 5) 135; 6) 45
Check result: Code word: FABRIC
Teacher note: Six items about the link between the average, the total and the number of data: the total from an average, a missing fourth number, the new average after a fourth mark, the age of a child who joins, the number taken away and the average of the two that stay. The pupil works with totals, not with averages of averages. The decoy numbers are what a pupil gets by adding or subtracting the averages, or by dividing by the wrong number. A box left blank gives no letter and the check cannot see it.

S2-12 (P6-32) Colour the Butterfly 2 stars

Topic: Volume and average

Syllabus outcomes: Area and Volume 2.1, Area and Volume 2.2, Area and Volume 2.3, Statistics 1.1, Statistics 1.2. Notes on the tag: Volume 2.1, 2.2, 2.3 and Statistics (volume of a cuboid or cube, the missing height or edge, average, total and number)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 27 = yellow, 36 = red, 48 = blue, 64 = green, 72 = orange
Check result: Tally: yellow 2, red 4, blue 2, green 4, orange 2; the parts in the same place on the left and on the right match.



Teacher note: Every part asks about the volume of a cuboid or cube (the volume from the edges, the height from the volume and the base area, a cube's edge or volume) or about an average (from a list, from a total, or the total from an average). Volumes are in cm³. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

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S2-13 (P6-33) Pie Chart 2 stars

Topic: Pie charts: reading percentages and finding amounts

Syllabus outcomes: Revisits Primary 4 Statistics, Revisits Primary 5 Percentage. Notes on the tag: Revisits Primary 4 Statistics (reading and interpreting pie charts) with Primary 5 percentage
Puzzle type: Code-breaker (pie chart). How the pupil checks: Code word (6 letters).
Answers: 1) 360; 2) 3; 3) 140; 4) 180; 5) 10; 6) 40
Check result: Code word: DONKEY
Teacher note: A pie chart with five sectors (percentages shown, one hidden). Six questions: the hidden percentage (the sectors add up to 100%), the number of pupils in a sector, a difference of two sectors, the number who did not choose one sport, the number in the hidden sector (which needs the first answer) and how many times one sector is another. The decoy numbers are what a pupil gets by using the percentage as the number, by adding or taking 10 or 5, or by using 100 minus the answer. A box left blank gives no letter and the check cannot see it.

S2-14 (P6-34) Net or Not 1 star

Topic: Nets: can it fold to make a cube?

Syllabus outcomes: Revisits Primary 4 Geometry. Notes on the tag: Revisits Primary 4 Geometry (identifying nets of a cube)
Puzzle type: Choice table (nets of a cube). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) Yes; 2) Yes; 3) No; 4) Yes; 5) No; 6) No
Check result: Word: FAMILY
Teacher note: Six nets of six squares: three fold to make a cube and three do not (a net with a block of four squares, a row that is too long, or two squares on the same side that would overlap). Each row has two answers, Yes and No, each with a letter. The letters of the right answers, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. The key shows which nets fold. A box left blank gives no letter and the check cannot see it.

S2-15 (P6-35) Taps and Rates 2 stars

Topic: Rate: the amount of one quantity for each unit of another

Syllabus outcomes: Revisits Primary 5 Rate, Revisits Primary 5 Volume. Notes on the tag: Revisits Primary 5 Rate (finding the rate, the total or the number of units); volume of liquid
Puzzle type: Code-breaker (rates). How the pupil checks: Code word (6 letters).
Answers: 1) 22; 2) 405; 3) 198; 4) 7; 5) 120; 6) 480
Check result: Code word: NUMBER
Teacher note: Six rate stories: the time to fill a tank at a rate in litres a minute, the number made in several hours, two taps open together, the amount still needed after some minutes, the rate of a pump and the number of words typed in a different time. The decoy numbers are what a pupil gets by multiplying instead of dividing (or the reverse), by using only one tap, or by forgetting the amount already filled. A box left blank gives no letter and the check cannot see it.

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S2-16 (P6-36) Mixed Numbers 1 star

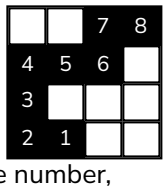
Topic: Fractions: mixed numbers, adding, subtracting and multiplying

Syllabus outcomes: Revisits Primary 5 Fractions. Notes on the tag: Revisits Primary 5 Fractions (adding and subtracting mixed numbers; multiplying a fraction by a whole number or a fraction)
Puzzle type: Ordering (mixed number answers). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: smallest answer first: S = 1_⅛, H = 2_⅙, A = 2_¾, D = 5_¼, O = 6_⅔, W = 6_¹¹⁄₁₂
Check result: Word: SHADOW
Teacher note: Six items: adding mixed numbers, subtracting mixed numbers (one needs renaming), a fraction times a whole number, a mixed number times a whole number, an improper fraction times a proper fraction and a story. The pupil works out all six answers and puts them in order, smallest first; the letters, read in that order, spell the word. The six answers differ by at least a quarter, so a pupil who makes a slip in one answer usually changes its place in the order. Every wrong order was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-17 (P6-37) Decimal Maze 1 star

Topic: Decimals: adding, subtracting, multiplying and dividing

Syllabus outcomes: Revisits Primary 4 Decimals. Notes on the tag: Revisits Primary 4 Decimals and Primary 5 (multiplying and dividing by 10, 100 and 1000)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every box equal to 2.4.
Answers: Route (row, column from top-left): (4,2) 0.024 × 100 -> (4,1) 2.9 − 0.5 -> (3,1) 2.3 + 0.1 -> (2,1) 1.75 + 0.65 -> (2,2) 19.2 ÷ 8 -> (2,3) 1.2 × 2 -> (1,3) 2.1 + 0.3 -> (1,4) 2.05 + 0.35
Check result: Route reaches FINISH through 8 boxes; every box on it equals 2.4.



Teacher note: A 4 by 4 grid of sixteen decimal sums: adding, subtracting, multiplying by a whole number, dividing by a whole number, and multiplying or dividing by 10 or 100. The route follows the boxes that equal 2.4. The wrong boxes have answers close to 2.4, such as a value ten times too big or too small, which a pupil gets by misplacing the decimal point. Two extra boxes that equal 2.4 are dead ends. Counting the boxes shows a line that takes a shortcut.

S2-18 (P6-38) GST and Discount 1 star

Topic: Percentage: GST, discount and interest

Syllabus outcomes: Revisits Primary 5 Percentage. Notes on the tag: Revisits Primary 5 Percentage (a percentage part of a whole; discount, GST and annual interest)
Puzzle type: Code-breaker (GST, discount, interest). How the pupil checks: Code word (6 letters).
Answers: 1) $90; 2) $20; 3) $54; 4) $436; 5) $36; 6) $45
Check result: Code word: TURKEY
Teacher note: Six items on percentage in money: the price with GST, a sale price after a discount, the amount of GST, a year's interest, the amount of a discount, and a discount found from the usual price and the price after it. The GST rate used is 9% and is printed on the page; the teacher may change it if the rate changes. The decoy numbers are what a pupil gets by adding 9 or 10 instead of 9%, by giving the discount instead of the sale price (or the reverse), or by being ten times out. A box left blank gives no letter and the check cannot see it.

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S2-19 (P6-39) Mixed Bag 2 stars

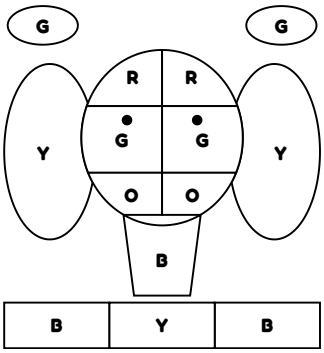
Topic: Mixed practice: percentages, fractions, ratio, average, roots, letters and volume

Syllabus outcomes: Fractions 1.2, Ratio 1.3, Algebra 1.4, Statistics 1.1, Area and Volume 2.5, Revisits Primary 5 Percentage, Revisits Primary 5 Volume (problem solving). Notes on the tag: Mixed revision: Percentage, Fractions 1.2, Ratio 1.3, Algebra 1.4, Statistics, Volume and revisited Primary 5 percentage
Puzzle type: Domino loop (mixed). How the pupil checks: Loop: the right half of each domino equals the left half of the next, all the way round to START; the letters in loop order spell a word (6 letters).
Answers: Loop from START: W ($\sqrt[3]{216}$ + 6 | average of 48, 41 and 58) -> O (average of 39, 50 and 58 | 10% of 40) -> M (3 ÷ ¾ | average of 13, 14 and 18) -> B (7 p + 1 when p = 2 | 50% of 48) -> A (larger share of 28 in the ratio 1 : 6 | 4 n + 2 when n = 9) -> T (2 t + 26 when t = 6 | 20% of 60)
Check result: Word: WOMBAT
Teacher note: Six dominoes with twelve halves. Each half is a short question with a whole-number answer: a percentage of a number, a fraction divided by a fraction, an average of three numbers, a larger share in a ratio, a root with a number added, an expression with a letter, or the volume of a cuboid. The right half of one domino has the same value as the left half of exactly one other domino, so the dominoes make one loop that comes back to START. The pupil finds the value of each half, follows the loop from START and writes the letters in loop order; they spell the word. Every wrong order was tested against a very large word list: none spells a word.

S2-20 (P6-40) Colour the Elephant 2 stars

Topic: Mixed Primary 6 practice

Syllabus outcomes: Revisits Primary 5 Percentage, Fractions 1.2, Algebra 1.4, Algebra 1.5, Ratio 1.3, Statistics 1.1. Notes on the tag: Mixed: percentage of a number (revisits Primary 5), Fractions 1.2, Algebra 1.4 and 1.5, Ratio 1.3, Statistics 1.1
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: 15 = yellow, 24 = red, 35 = blue, 48 = green, 60 = orange
Check result: Tally: yellow 3, red 2, blue 3, green 4, orange 2; the parts in the same place on the left and on the right match.

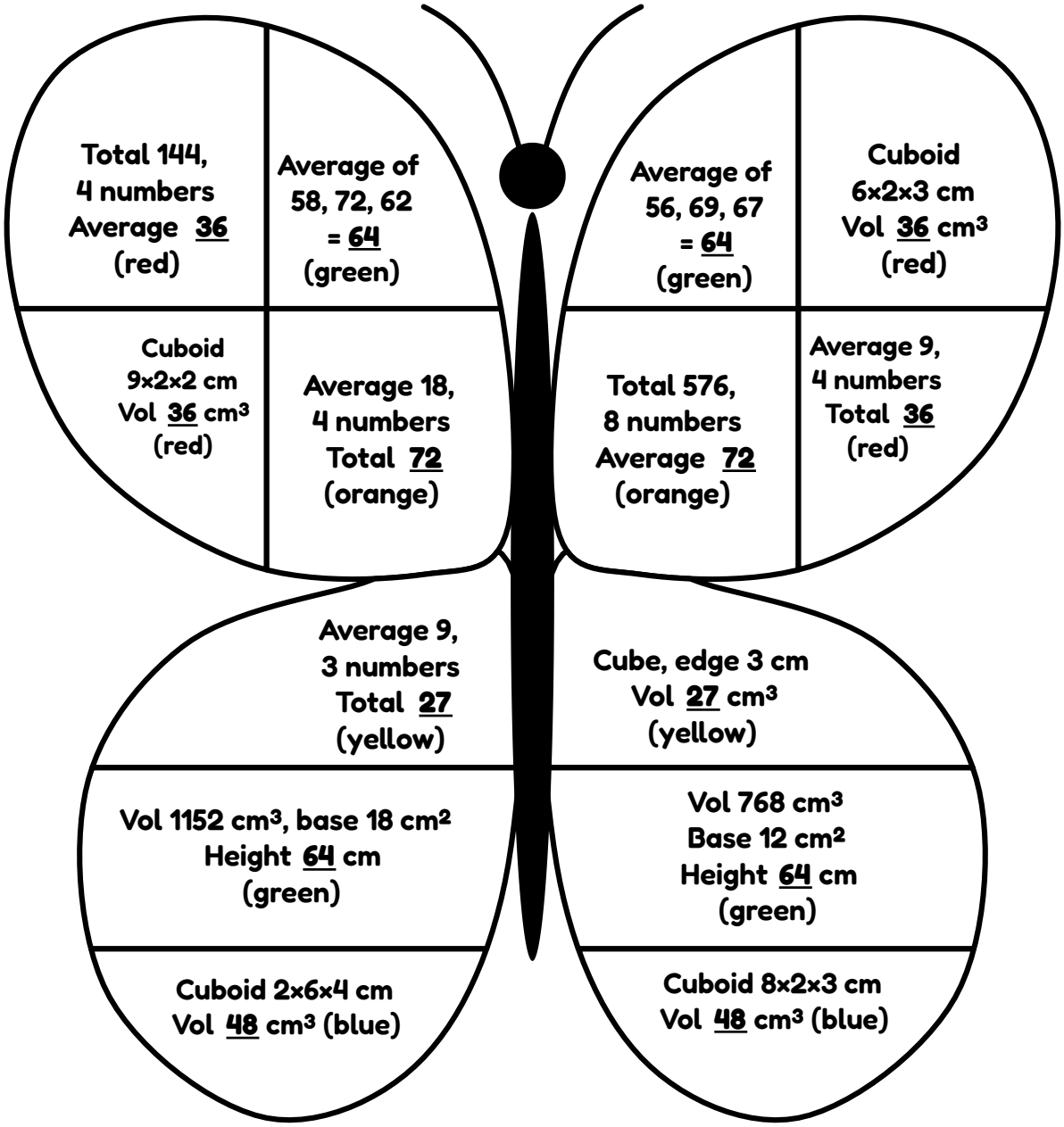


Teacher note: Every part asks a mixed Primary 6 question with a whole-number answer: a percentage of a number, a whole number divided by a fraction, the value of an expression with a letter, an equation with a letter, a share in a ratio, or an average. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

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Answers: S2-12 (P6-32) Colour the Butterfly

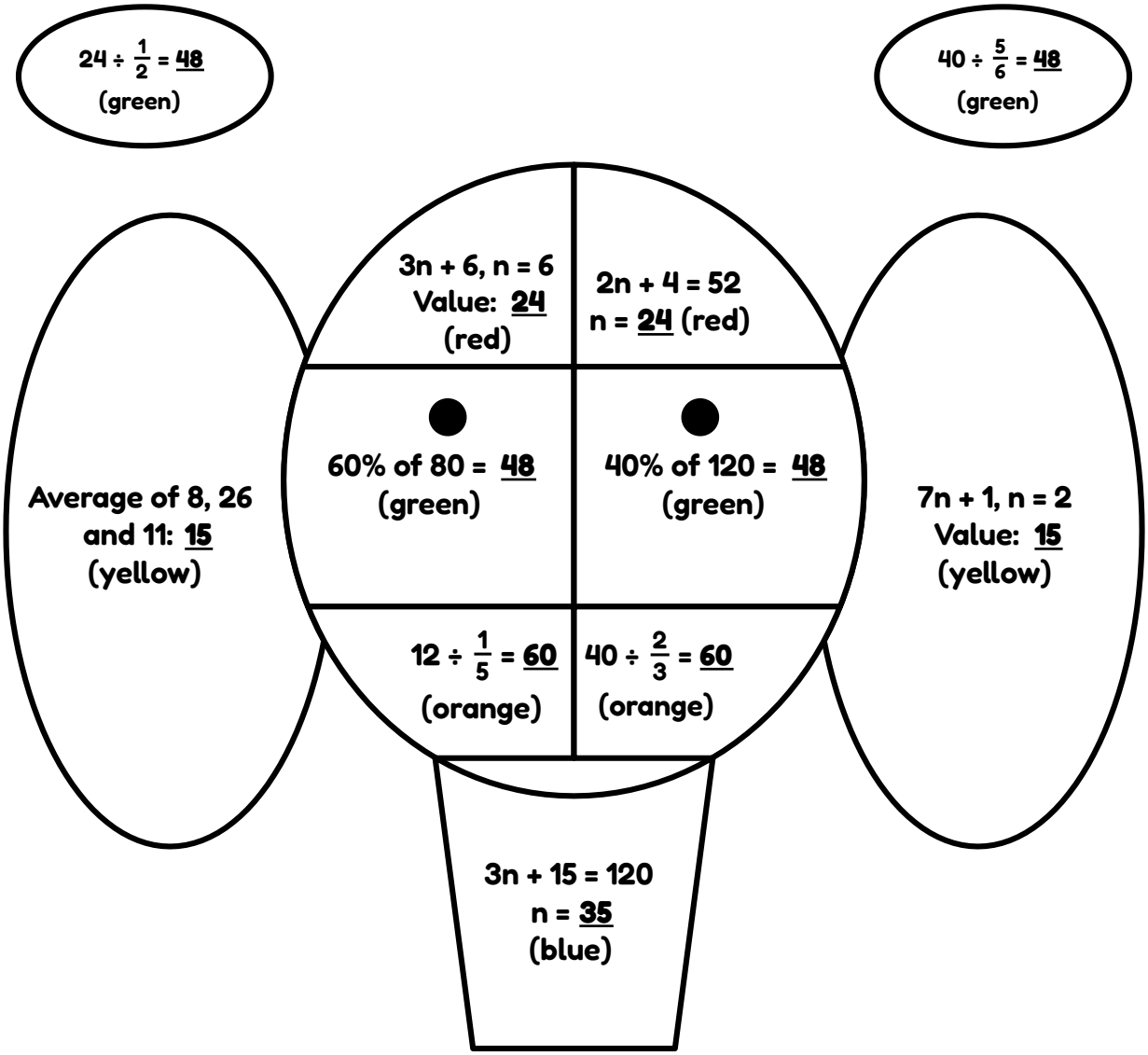
Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 27 = yellow, 36 = red, 48 = blue, 64 = green, 72 = orange.



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Answers: S2-20 (P6-40) Colour the Elephant

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 15 = yellow, 24 = red, 35 = blue, 48 = green, 60 = orange.



2n + 29, n = 3 Value: 35 (blue)	Share 21 in the ratio 5 : 2. Larger: 15 (yellow)	8n + 3 = 283 n = 35 (blue)

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